



Now That You've Downloaded Some StreetLight Data, What Should You Do First? Data Representativeness and Expansion Considerations

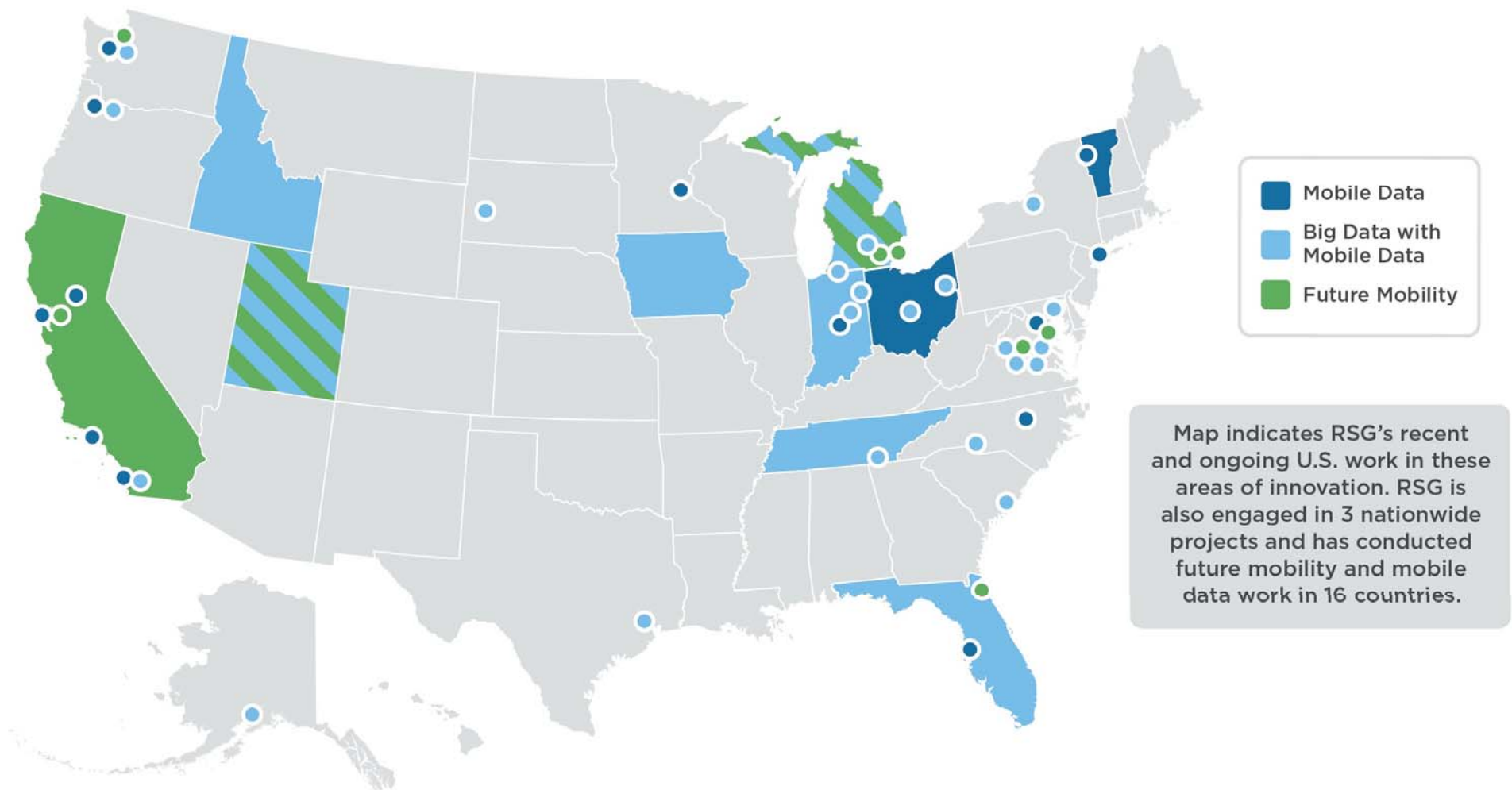
Vince Bernardin, PhD, RSG

September 8, 2017

Quick Recap of last December



RSG Experience





The Power of Big Data

The Power of Big Data

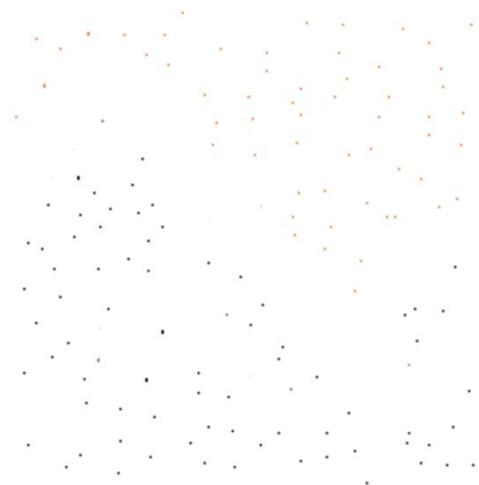
TN STATEWIDE DATA

- Combined household survey
 - NHTS + 4 MPOs
 - 10,344 households
- AirSage and ATRI datasets
- Trip Table (OD pairs)
 - Total: 12,744,900
 - Survey: 39,782 0.3%
 - AirSage: 3,355,539 26.3%

CHATTANOOGA DATA

- 2010 household survey
 - 1,502 households
- AirSage and ATRI datasets
- Trip Table (OD pairs)
 - Total: 529,984
 - Survey: 8,350 2.0%
 - AirSage: 182,742 34.5%

Can you recognize the pattern based on $<2\%$?



How about based on >25%?



Big Data allows us to see the Big Picture



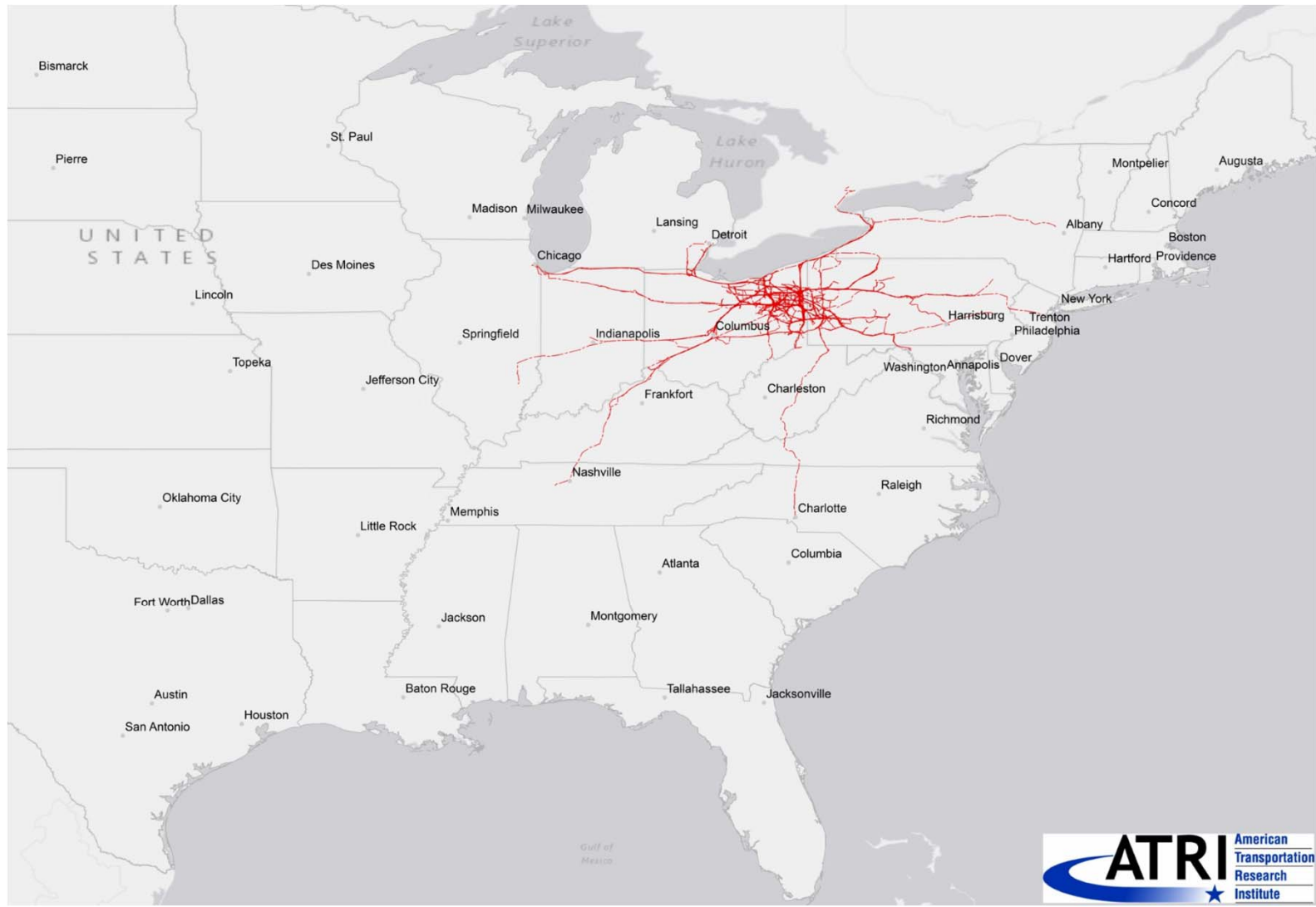
But what about this 25%?



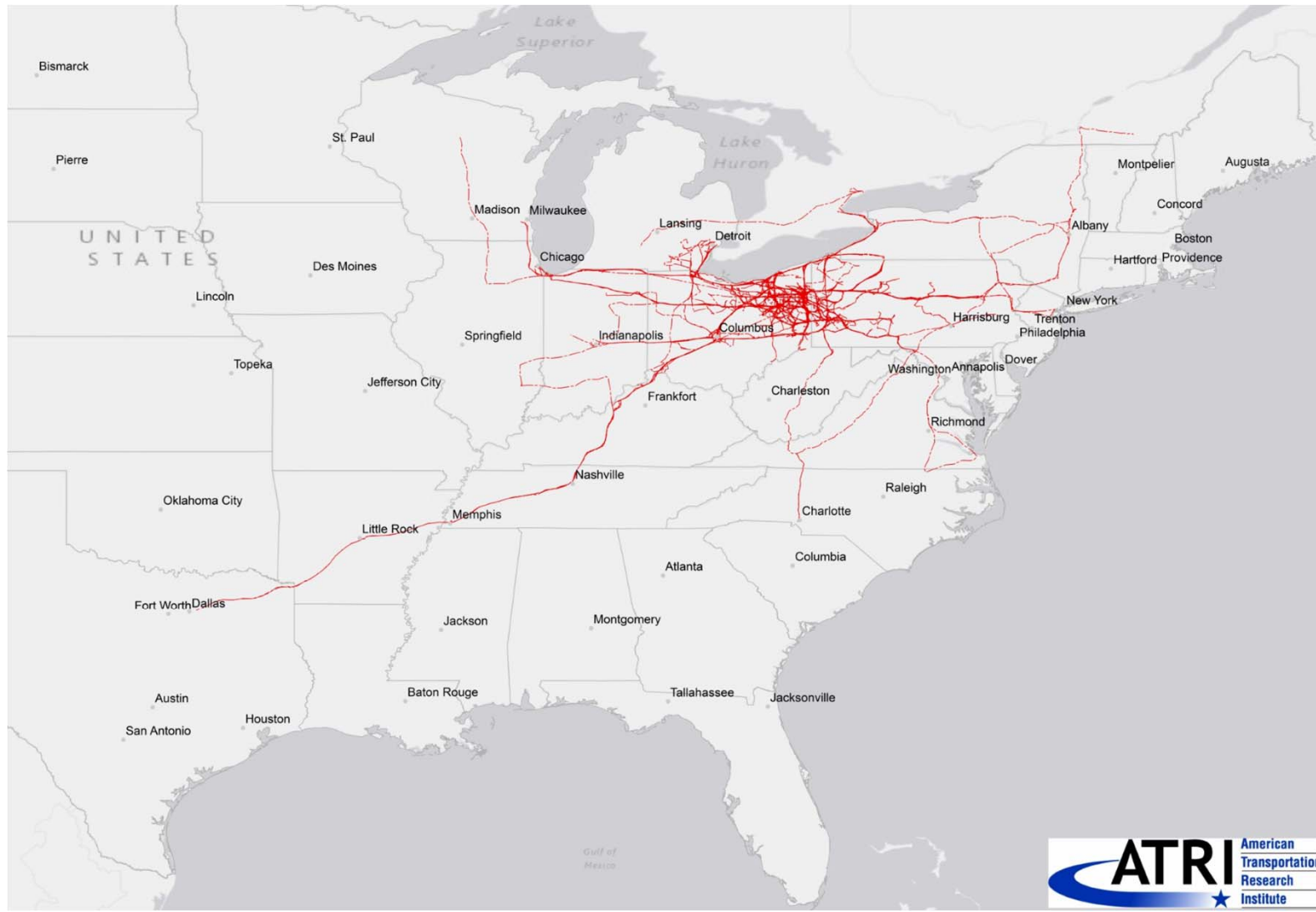
US 30 Study Area



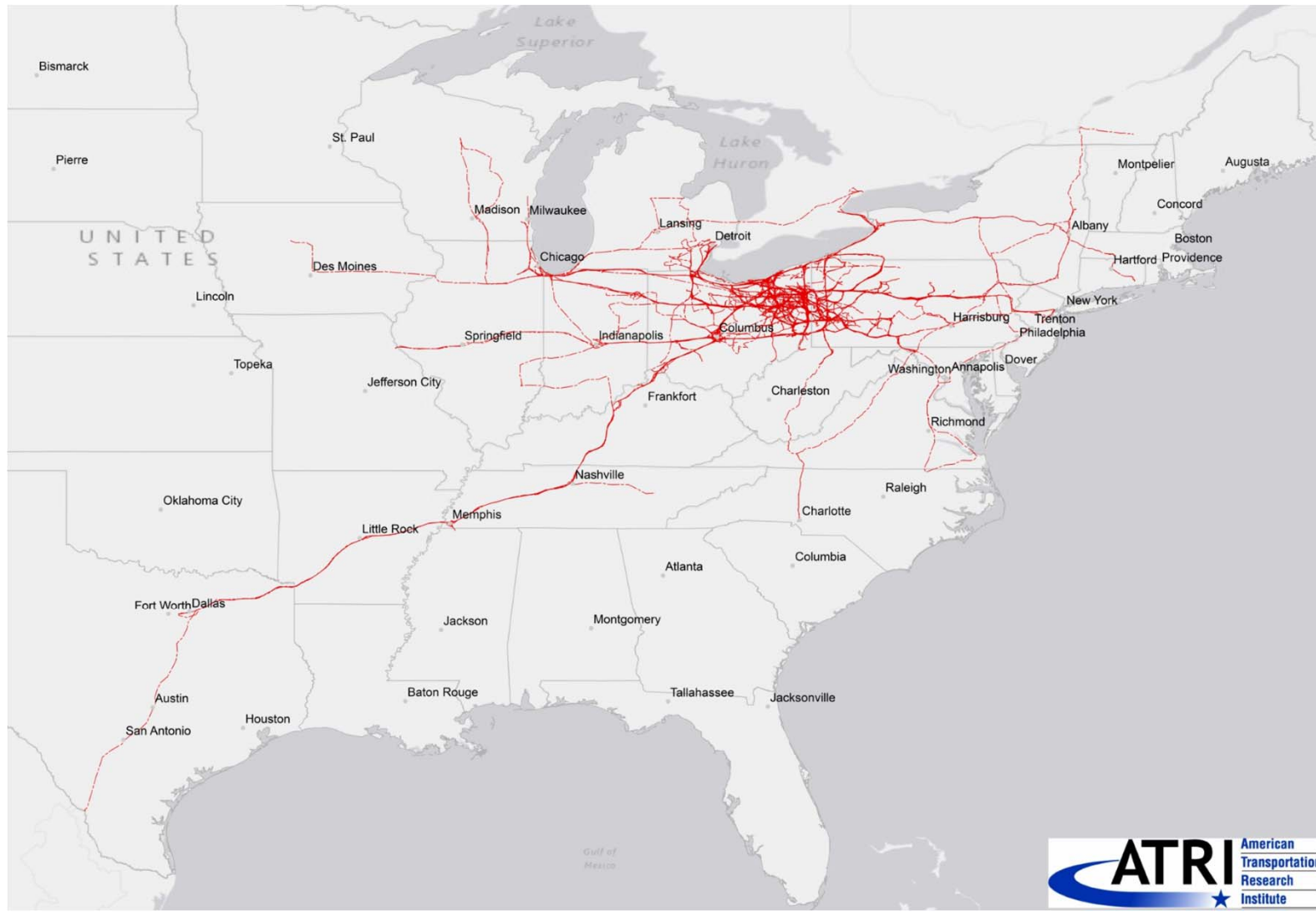
Trucks using the US 30 corridor – after 1 Day



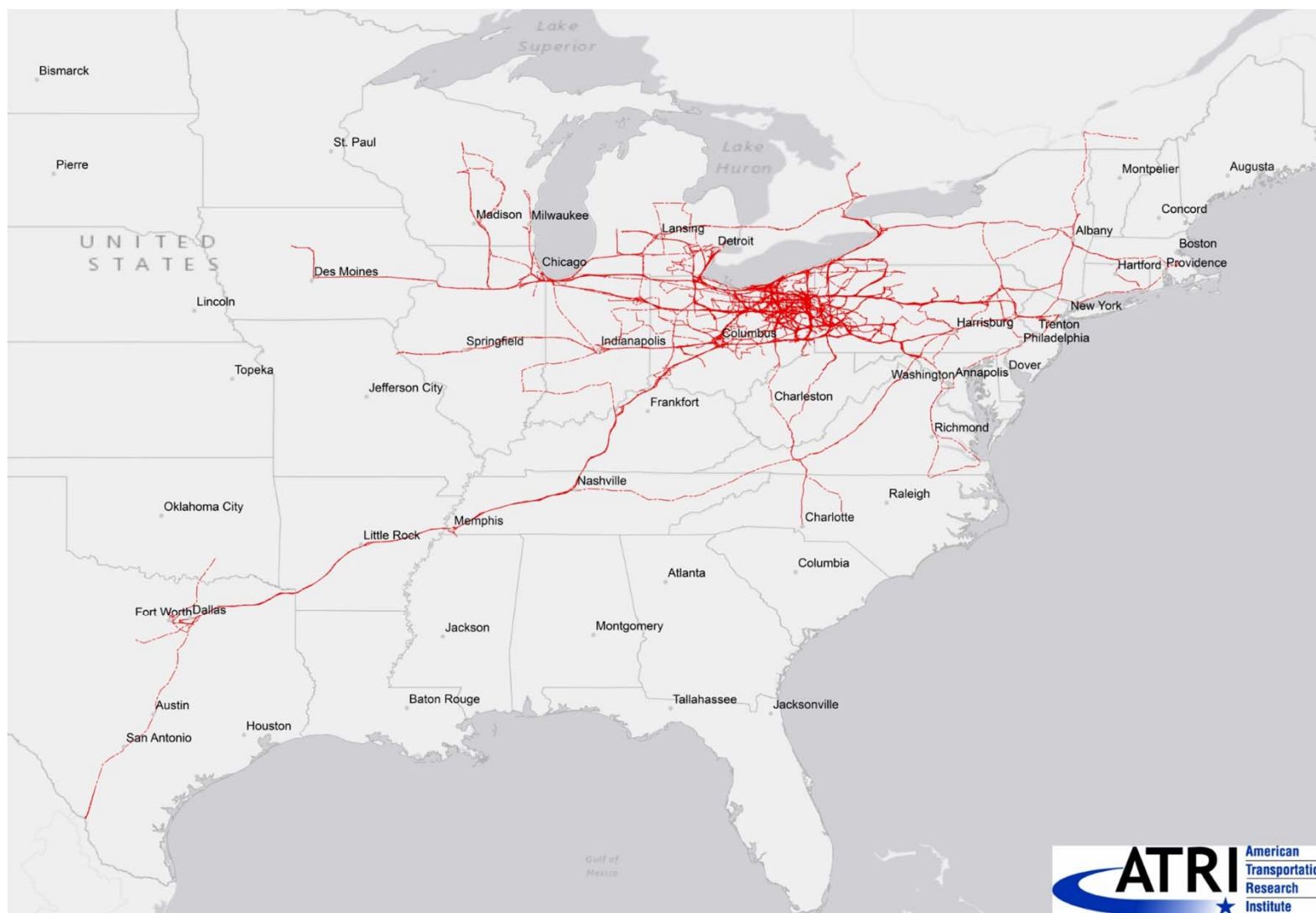
Trucks using the US 30 corridor – after 2 Days



Trucks using the US 30 corridor – after 3 Days



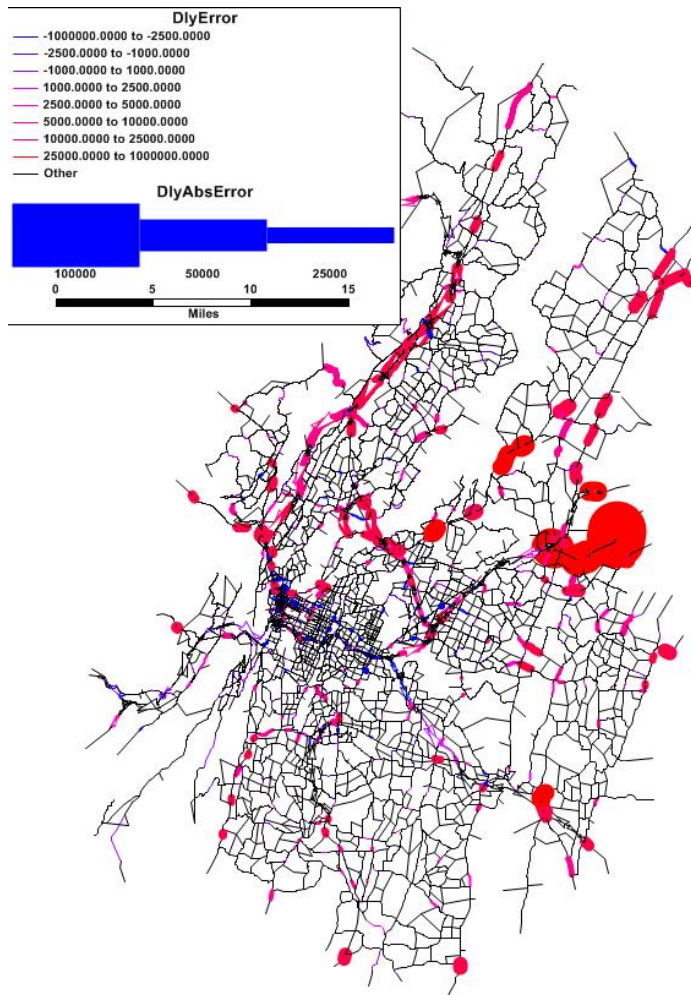
Trucks using the US 30 corridor – after 5 Days





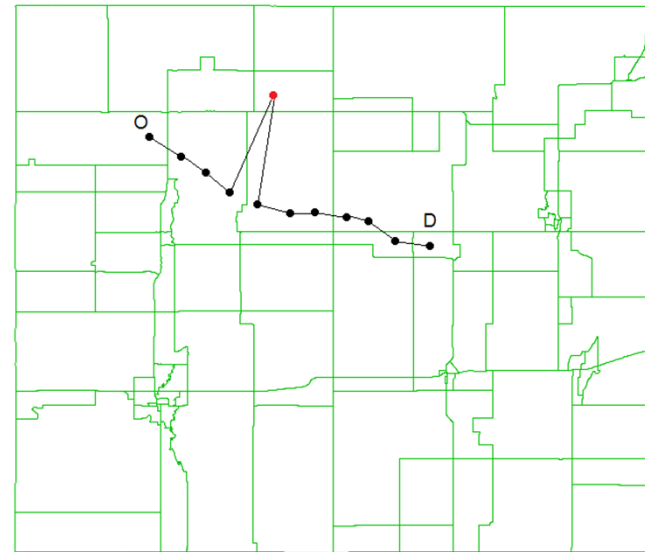
The Limitations of Big Data

Cleaning Required



- Filtering / cleaning

- Needs vary by data source – but all need it
- GPS jumps/blips and equivalent
- Missing data



Not Representative

- Big Sample NOT Random Sample
 - Locational biases, holes
 - Trip length / duration biases
 - Not corrected by penetration-based expansion





Model Results with Big Data

To/From TN Trip Distribution

DISTRICT-TO-DISTRICT COMPARISON

- Generally good agreement
- District level origins & destinations all within 10%, most within 3%
 - Smoky Mtns not attracting enough to/from Knoxville
- District level ODs all within 4% except within Nashville - Northcentral

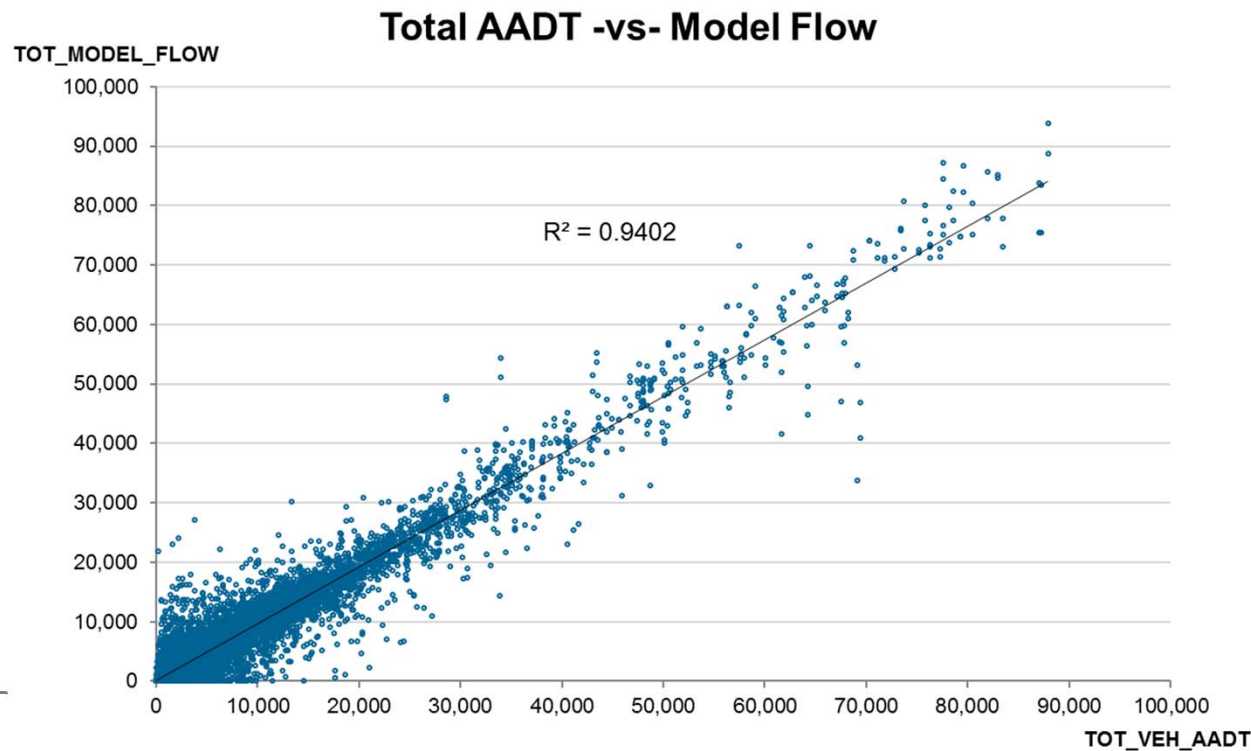
Relative Percentage Difference (Model Version 3 vs AirSage) I-E & E-I Trips

Internal districts	External districts							Total
	Northwest	North Atlantic	Northcentral	Carolinas	Alabama-Gulf	Southwest	Georgia-Florida	
Tri-Cities	0.4%	0.1%	0.8%	3.6%	0.0%	0.2%	0.3%	5.3%
Knoxville	0.5%	-2.6%	-1.2%	-1.7%	-0.7%	0.3%	-2.0%	-7.3%
Chattanooga	0.0%	-0.1%	-0.5%	-0.4%	-1.1%	0.1%	2.7%	0.8%
Cookeville	0.0%	-0.2%	0.9%	-0.3%	-0.1%	-0.1%	-0.2%	0.0%
Lynchburg	-0.4%	0.1%	0.4%	0.0%	0.7%	-0.1%	-0.4%	0.2%
Nashville	-0.7%	-0.3%	6.6%	-0.8%	-3.6%	-2.3%	-2.0%	-3.1%
Jackson	0.0%	0.1%	0.6%	0.0%	0.0%	-1.9%	0.0%	-1.2%
Memphis	0.5%	0.3%	0.8%	0.1%	-0.1%	3.4%	0.3%	5.2%
Total	0.3%	-2.6%	8.4%	0.5%	-4.9%	-0.4%	-1.3%	0.0%

Assignment Validation

- Great fit - One of best statewide models in the country
- Used ODME – with constraints, (some other statewide models do to)

VOLUME RANGE	RMSE	TDOT TARGET
< 5,000	102.1%	101.4%
5,000 to 10,000	35.6%	56.3%
10,000 to 20,000	22.0%	51.4%
20,000 to 30,000	16.4%	35.7%
30,000 to 40,000	14.8%	32.0%
> 40,000	11.1%	12.2%
All	36.6%	60.0%



Total Daysim Trip Table vs. AirSage

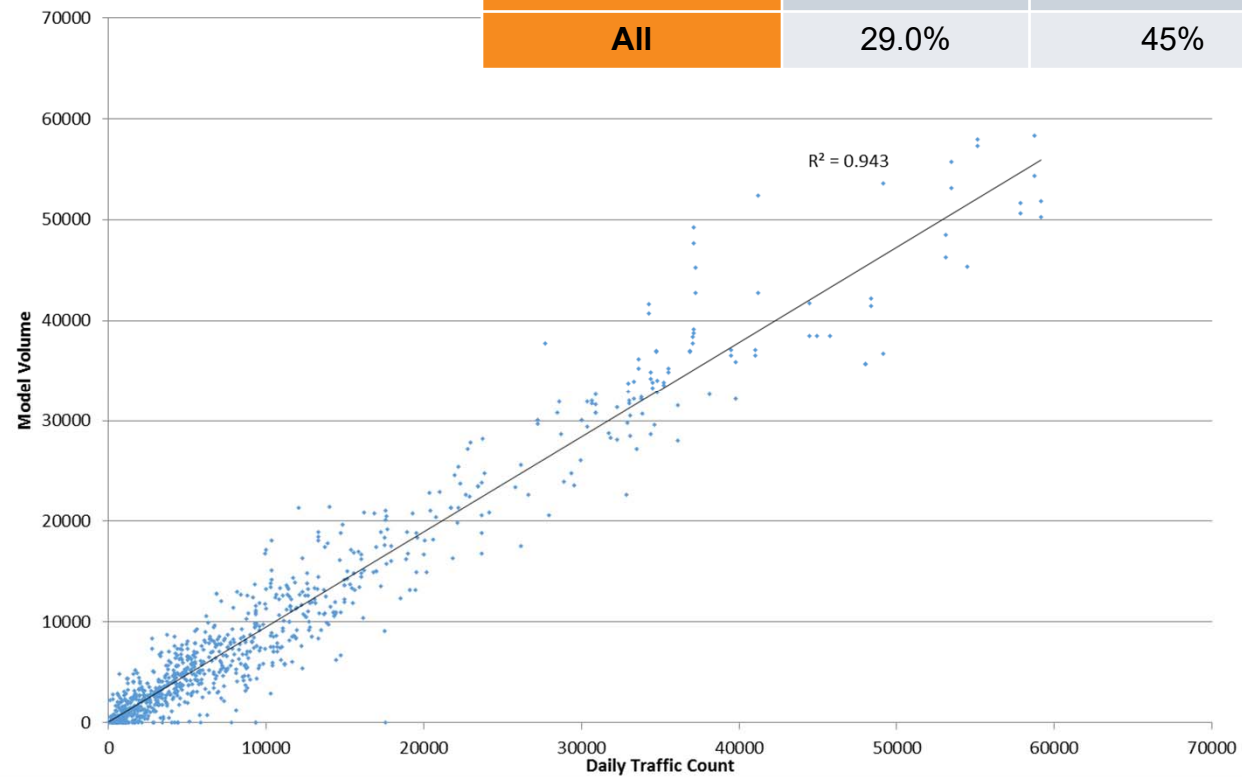
- Daysim vs. AirSage
 - Very good agreement – **10.5% RMSE**
 - All cells within +/- 1%
 - All residence/work Super Districts within +/-2.5%

Origin SuperDistrict	Destination Super District												Grand Total
	1	2	3	4	5	6	7	8	9	10	11	12	
1	0.5%	0.2%	-0.1%	0.0%	0.0%	-0.1%	-0.2%	-0.1%	0.0%	0.0%	-0.1%	-0.2%	0.0%
2	0.3%	0.0%	0.2%	0.0%	0.1%	0.0%	0.0%	0.1%	0.1%	0.0%	0.0%	-0.1%	0.7%
3	-0.1%	0.1%	0.0%	-0.1%	-0.2%	0.0%	0.1%	0.1%	0.0%	0.0%	0.0%	-0.1%	-0.1%
4	0.0%	0.1%	-0.1%	0.0%	0.0%	0.0%	0.1%	0.1%	0.0%	0.0%	0.0%	0.0%	0.4%
5	0.1%	0.1%	-0.1%	0.0%	0.2%	0.1%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.5%
6	-0.1%	-0.1%	0.1%	-0.1%	0.1%	0.0%	0.1%	-0.1%	0.1%	0.0%	0.0%	0.0%	0.0%
7	0.0%	0.0%	0.2%	0.1%	0.1%	0.0%	0.2%	0.0%	0.1%	0.0%	0.0%	0.1%	0.7%
8	0.0%	0.1%	0.1%	0.1%	0.0%	-0.1%	0.1%	0.0%	-0.2%	0.0%	0.0%	0.0%	0.2%
9	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%	0.3%	0.0%	0.0%	0.0%	0.2%
10	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.1%	0.0%	0.3%
11	0.0%	0.0%	0.0%	-0.1%	0.0%	0.0%	-0.1%	0.0%	0.0%	0.1%	-0.1%	-0.3%	-0.5%
12	-0.2%	-0.3%	-0.1%	-0.2%	0.0%	-0.1%	-0.2%	-0.1%	-0.1%	0.0%	-0.3%	-0.7%	-2.4%
Grand Total	0.5%	0.2%	0.2%	-0.2%	0.4%	-0.3%	0.4%	0.1%	0.3%	0.3%	-0.5%	-1.3%	0.0%

Assignment Validation

- Great fit!
 - Better than old model
 - Far exceeds TDOT standards
 - No ODME, only screenline factoring

VOLUME RANGE	RMSE	TDOT MAXIMUM
< 5,000	62.1%	100%
5,000 to 10,000	37.9%	45%
10,000 to 15,000	28.0%	35%
15,000 to 20,000	22.7%	30%
20,000 to 30,000	15.7%	27%
30,000 to 50,000	14.1%	25%
50,000 to 60,000	9.9%	20%
All	29.0%	45%





Data Validation and Cleaning

Tips for Clean Streetlight Data Queries

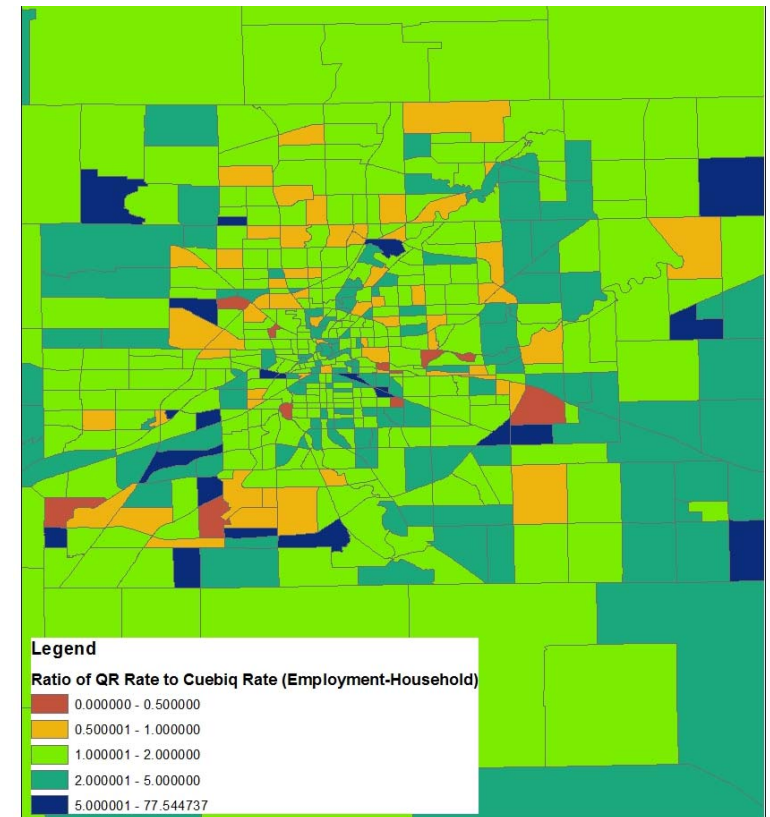
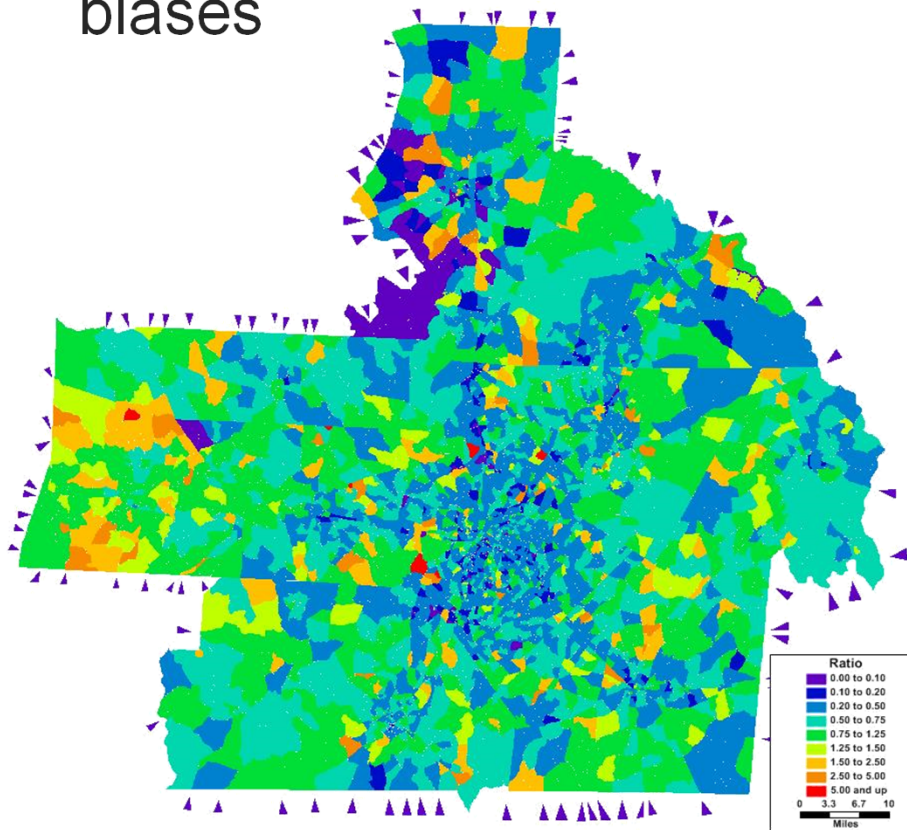
- Minimize data cleaning on the front end
 - For GPS external station data queries – use two directional pass-through zones per external station
 - Separate queries of EE and EI/IE will result in double-counting
 - Be careful/intentional about the treatment of intermediate stops on EE trips / whether you want one EE trip or an EI and IE
 - For LBS external catchment/buffer zones, use length based on corridor speeds (further for interstates)
 - For LBS queries around water, buffer out some distance (e.g., 50 or 100 m) into the water – but consider leaving gaps between zones in wide rivers

Trip Rate Comparisons

- Start by examining the marginals of the OD matrix and comparing them to zonal SE data
 - Examining trip rates can identify various data problems
 - Coverage holes (more of an issue for cellular data, but can also arise from line-of-sight issues such as urban canyons for GPS data)
 - Sampling issues (generally more of an issue for commercial vehicle data because of the lumpiness of the sample)
 - SE data issues (Oh, right that place did close! Oh, yeah, there's a new subdivision there!)
 - Internal validity checks examine outliers within the dataset (e.g., zones with trip rates $<$ or $>$ 2 SD from the mean)
 - Comparisons against modeled or quick-response P+As often is even more helpful

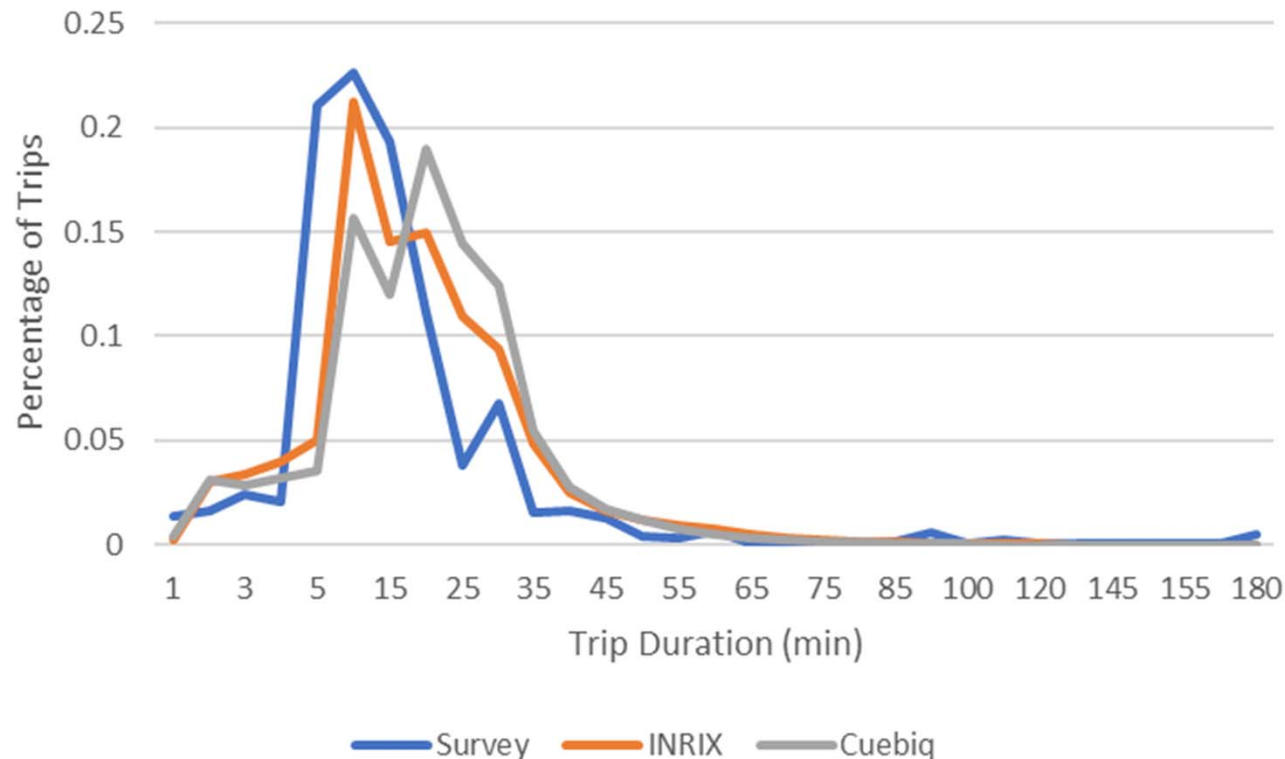
Trip Rate Comparisons

- Area type biases can be complicated and result from a combination of observation bias and trip/activity duration biases



Trip Length Comparisons

- Not always the most important or useful, but generally good practice to compare trip lengths where possible, too





Data Expansion

Overview of Methods

- Aware of 8 methods currently in use, and new methods being actively being researched
- Most robust expansion schemes combine several methods
 - SE data based and simple scaling to counts are among most commonly used and most commonly used alone
 - But these cannot correct for trip/activity duration biases
- Group methods first by type of control data used
 - Then subdivide count-based methods based on single/multiple factors, network based / not, parametric / non-parametric

Taxonomy of Expansion Methods

- SE Data Methods
 - **Market Penetration (Residence-based)**
 - **Trip Generation-Based**
- Traffic Count Methods
 - **Simple Scaling to Counts**
 - Multi-factor Scaling
 - Non-Assignment-Based
 - **Iterative Proportional Fitting to Counts (Frataring)**
 - **Iterative Screenline Fitting / Matrix Partitioning**
 - Network Assignment-Based
 - Nonparametric (ODME)
 - » **Direct ODME**
 - » **Indirect ODME**
 - **Parametric Scaling to Counts**
- *Trace Data Methods*

SE Data Methods

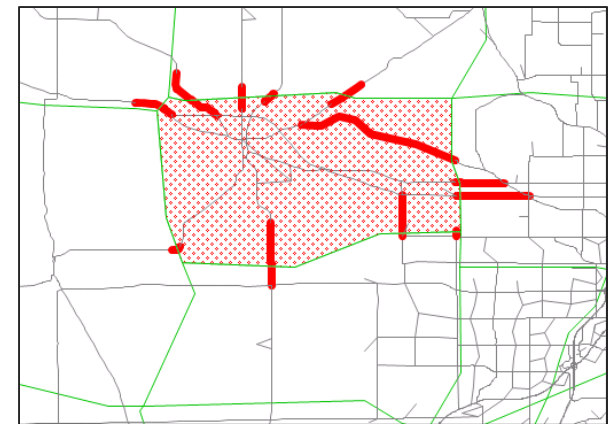
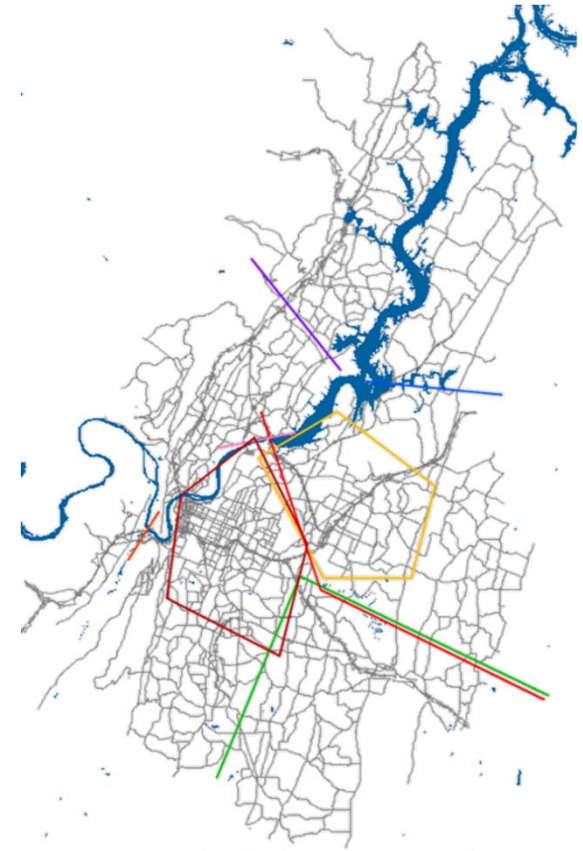
- Market Penetration-based
 - Requires device ID persistence to impute residence location
 - Not currently viable for GPS datasets
 - Compare resident devices per area to population to compute expansion factors by device residence areas
 - Good for addressing demographic biases, not for duration bias
- Trip Generation-based
 - Does not require residence imputation/ID persistence
 - Compares trips to/from zone to estimated trip to/from zone to estimate expansion factor
 - May be better for data validation than data expansion

Simple Count-based Methods

- Simple Scaling to Counts
 - Use a single expansion factor to minimize average loading error
 - Usually done via assignment but can be done with map-matching for data with sufficient locational precision (GPS, some LBS)
 - Almost always used as part of / in combination with other more complex count-based methods
 - Sometimes explained in terms of vehicle occupancy but this is only one of several effects that can be captured/reflected
- Iterative Proportional Fitting to Counts (Fratar)
 - Requires counts into/out of zone
 - Commonly used for expanding external stations
 - Also sometimes for airports and other special generator zones

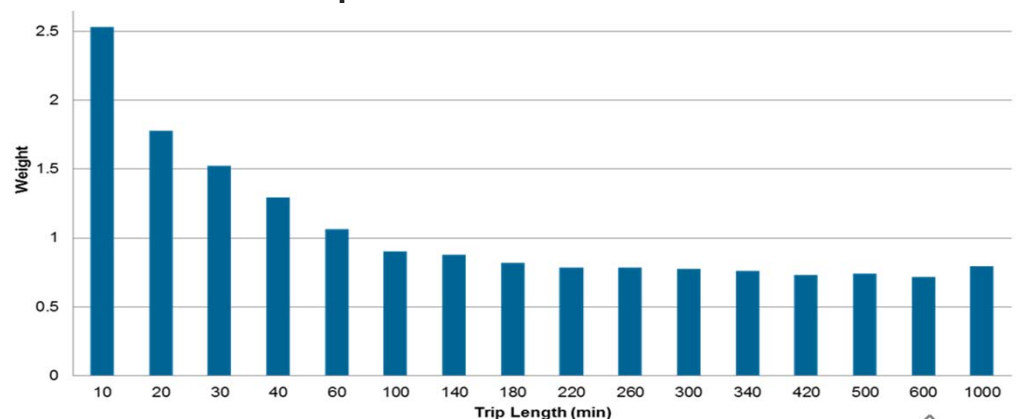
Iterative Screenline Fitting (ISF)

- Loop over screenlines
 - Uses screenlines which partition region into two sets of zones – which partition the OD matrix into quadrants
 - Diagonal quadrants receive factor of 1
 - Off-diagonal quadrants receive factor based on ratio of weighted total counts to aggregated OD trips
 - Weight based on number of screenlines each count is on, etc.
 - Average new factors from this screenline with prior expansion factors



Non-Parametric Assignment-based Methods

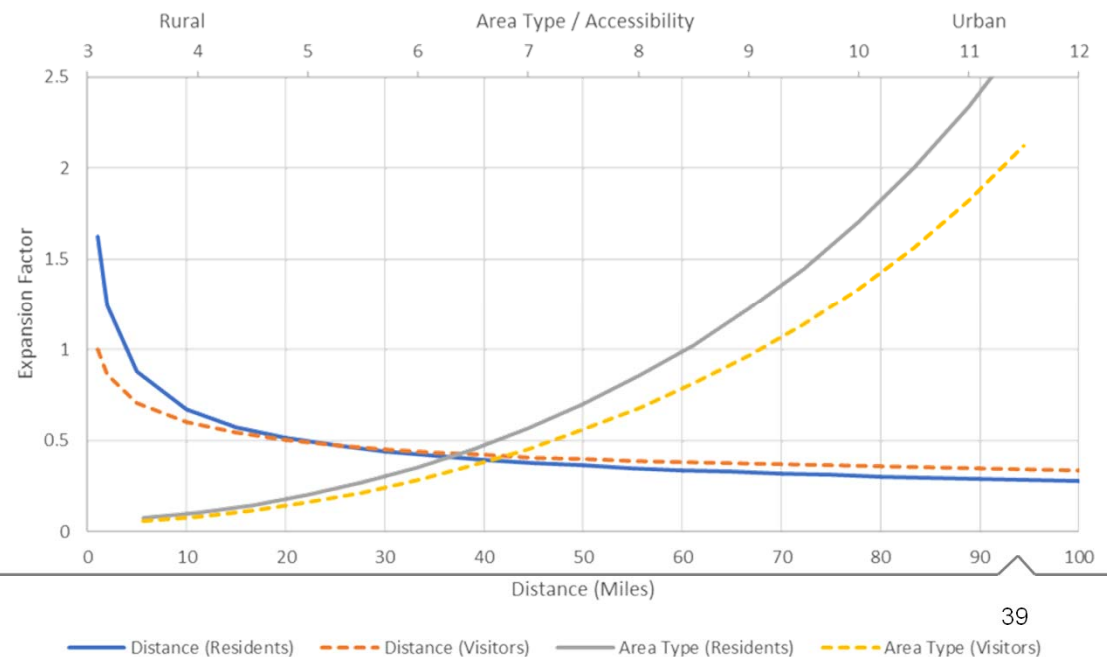
- Direct ODME
 - OD/cell-specific expansion factors (lots)
 - Beware of over-fitting to counts!
 - Many different ODME methods, **important** to use one that either minimizes error with respect to both counts and the original ODs or that minimizes error with respect to counts but only within certain constraints (e.g., -50% and +200%) – easier if ODME done after other methods
 - Should measure difference / distance from original to output OD flows (e.g., MAE, MAPE), not just compare TLFDs
 - Relatively easy to do but difficult to interpret / understand
- Indirect ODME
 - Analyze results of ODME to create simpler set of expansion factors based on distance, regions, etc.



Parametric Scaling to Counts

- Uses assignment within a larger framework to estimate / calibrate parameters for an expansion factor function
- Terms often include
 - Distance
 - Area type or accessibility
 - Intradistrict / intrazonal
 - Adjacency
- Estimation is NP-Hard
 - Mixed success with genetic algorithm
 - Mixed success with regression on ODME
 - Manual calibration

Charlotte Parametric Expansion Factor Curves

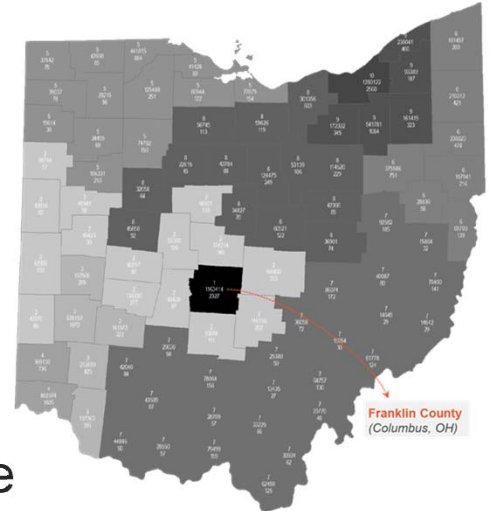




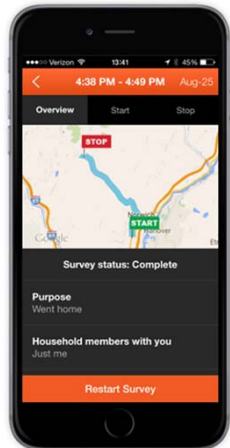
Disaggregate Trace Auditing

ODOT's HH Survey with rMove

- On-going Household Survey for Ohio
 - 1/10th of Ohio collected every year for ten years
 - About 75% collected by smartphone app, rMove
 - Mostly using their own phones, but some using loaners
 - These HH participate for 7 days
 - Subset of smartphone group doing 6 month long distance survey
 - About 25% collected online
 - These HH participate for 1 day



rMOVE
by RSG



	Group 1	Group 2	Group 3	Group 4
How many HH members age 16+ have smartphones?	All (100%)	Some	None (0%)	None (0%)
Given Choice for Participation	No	No*	Yes*	Yes
Study Participation Method	100% use own smartphones	mix of owned and borrowed phones	100% borrow smartphones	non-GPS (online diary)
Number of Days of Participation	7 Days	7 Days	7 Days	1 day
Total Count Recruited HHs	4497	506	619	1739
Total Percent Recruited HHs	61%	7%	8%	24%
Total Count Completed HHs	1751	176	359	792
Total Percent Completed HHs	57%	6%	12%	26%

* HHs recruited between March-May 2017 were not offered the option to borrow phones due to capacity limitations; HHs where some or no members owned smartphones were automatically assigned to the online diary during this time

Comparison of Cuebiq and rMove for TMIP

Better Methods. Better Outcomes.

- TMIP interested in deep dive to understand fundamentals of new big data sets
 - University of Washington looking at cellular and GPS data
 - RSG looking at LBS data from Cuebiq
 - ODOT agreed to allow use of rMove data
- First known disaggregate audit of passive data traces
 - High level of confidence in rMove
 - Trace verified by traveler, screened by AI, and if questionable reviewed by human analyst
 - LBS exiting, but new and we want to understand it better
 - Also, could help identify any remaining issues with rMove traces
 - Also, first step toward new data fusion techniques

Summary of Cuebiq data for Franklin, Co.

**Counts for Cuebiq data limited to a Wednesday in April 2017:
Includes users who passed through Franklin County, OH**

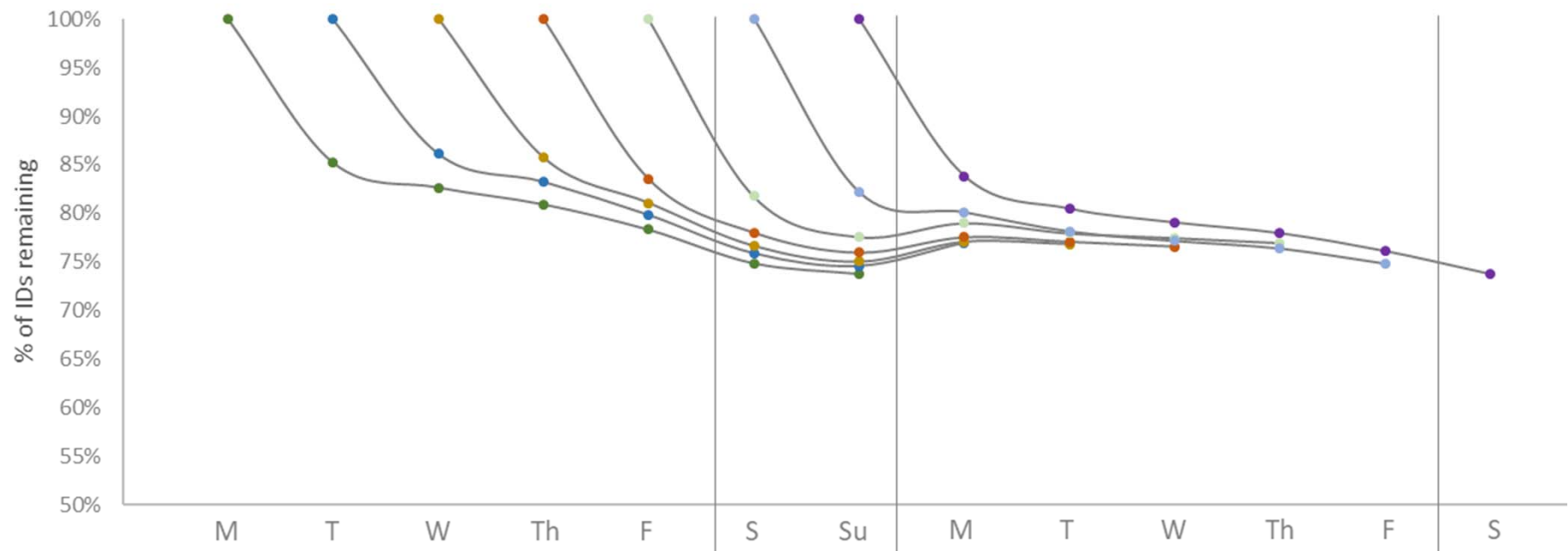
		Total
Total	Devices	95,697
	Points	12,128,310
Median time between points		2.5 minutes
% of points	IOS	54%
	Android	46%



Devices and points by accuracy

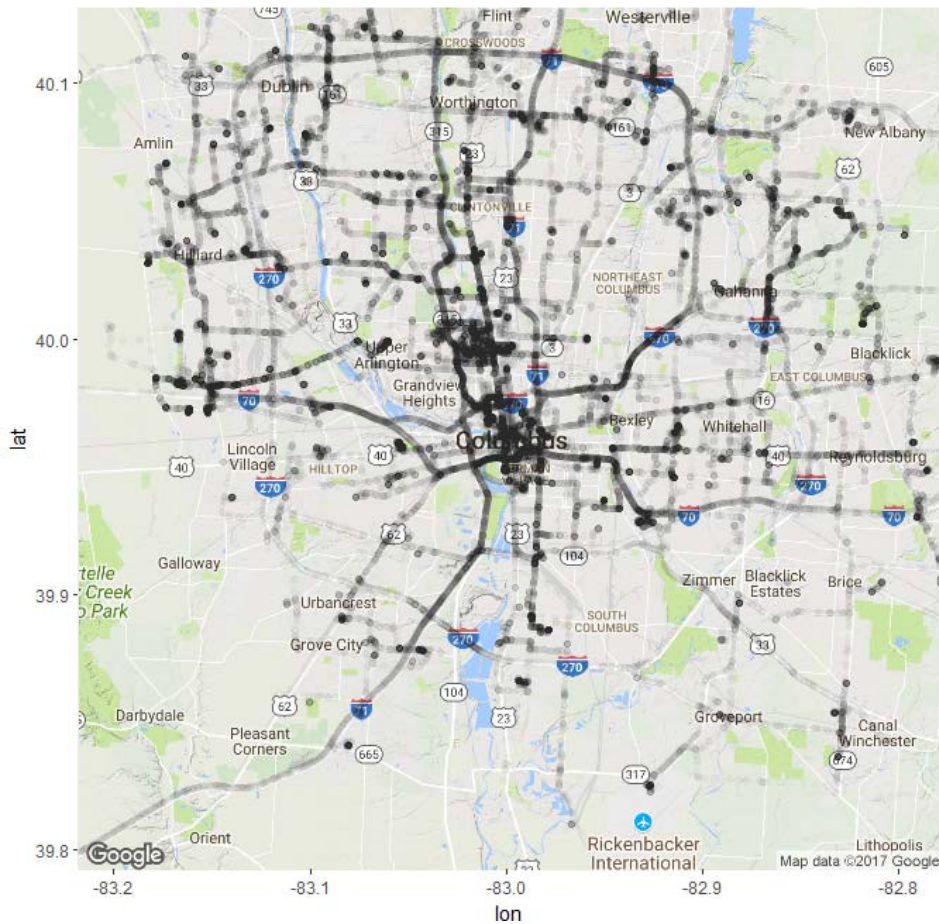
		Accuracy					
		1 m	< 10 m	< 50 m	< 100 m	< 1000 m	ALL
Total	Devices	33,716	48,300	91,915	94,123	95,300	95,697
	Points	1,037,151	3,553,114	8,093,624	9,863,018	10,786,426	12,128,310
% of points	IOS	100%	98%	66%	61%	59%	54%
	Android	0%	2%	34%	39%	41%	46%

Cuebiq ID Persistence over Time

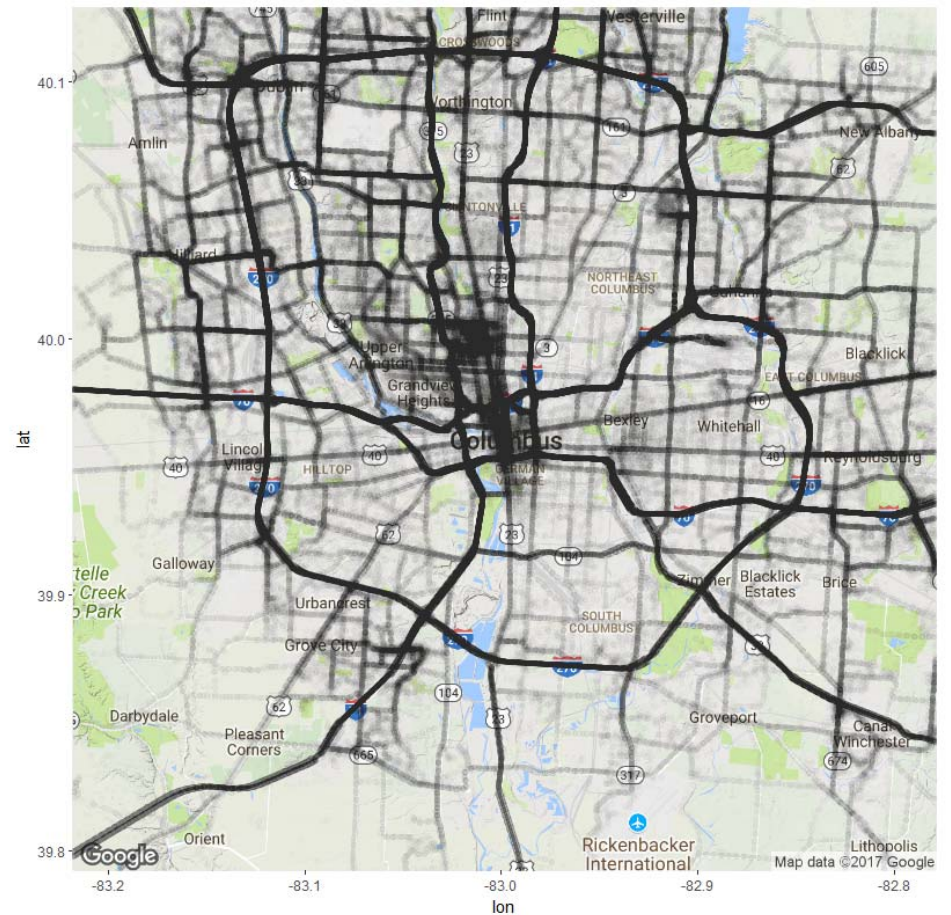


- ID persistence is quite strong
 - 70% to 80% of IDs are still active after a week
- Persistence is impacted by the weekend
 - Phone and app usage may vary by day of week

Mapping rMove vs. Cuebiq Locations

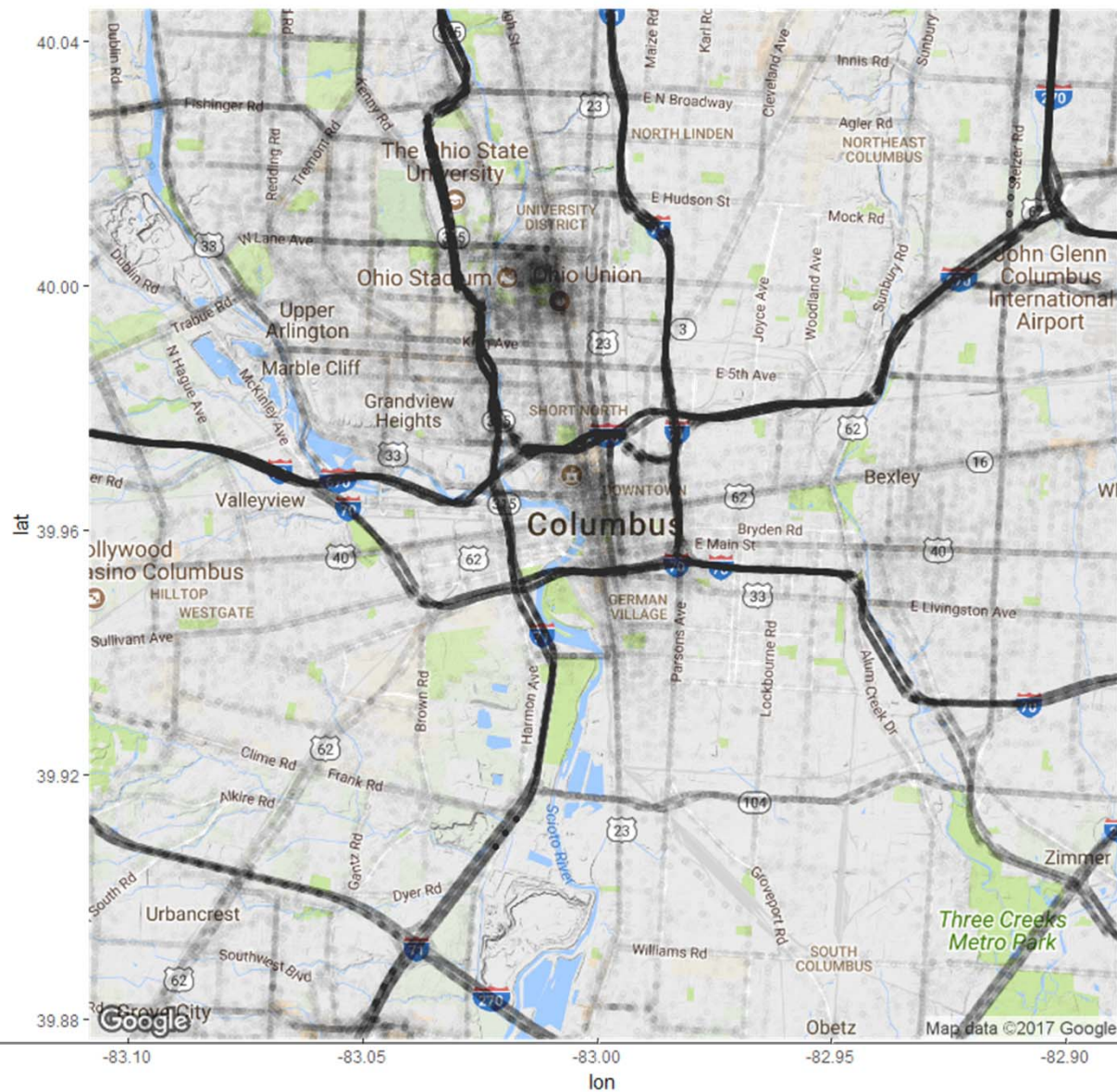


rMove – Wednesday in April



Cuebiq – Wednesday in April

Cuebiq Locations – Detail



Aggregate Cuebiq to rMove Comparison

Summary of trip data from a Wednesday in April 2017, based on initial trip inference algorithm, compared to unweighted rMove data from the same day

	Cuebiq inferred trips	rMove trips*
Number of Trips	76,780	1,187
Number of Devices	17,545	222
Trips per Device	4.38	5.34
Median Distance	3,462 m	4,208 m
Mean Distance	11,324 m	10,354 m

**includes incomplete data not delivered to ODOT*

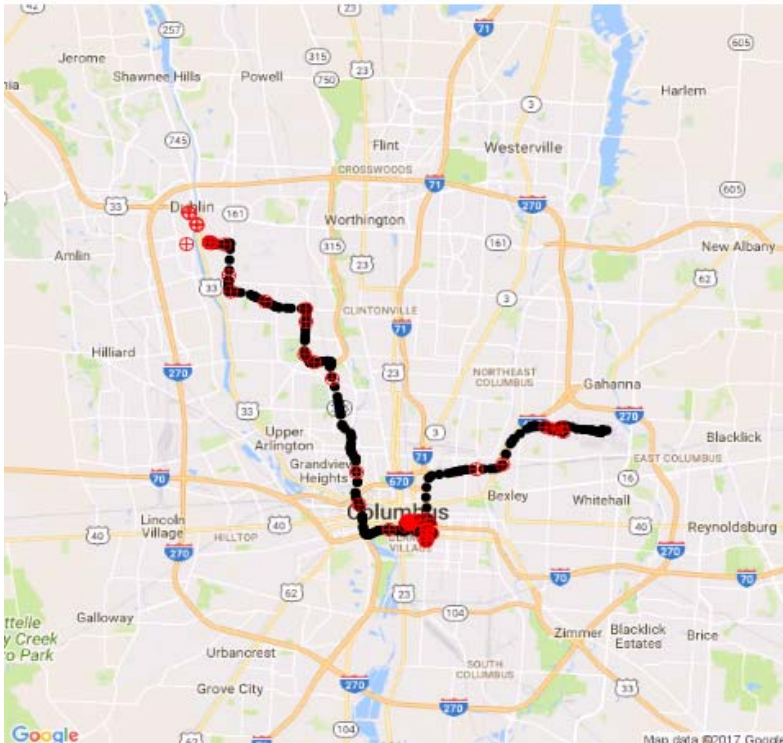
What are the
20% missing
from LBS?

Matching users in rMove/Cuebiq

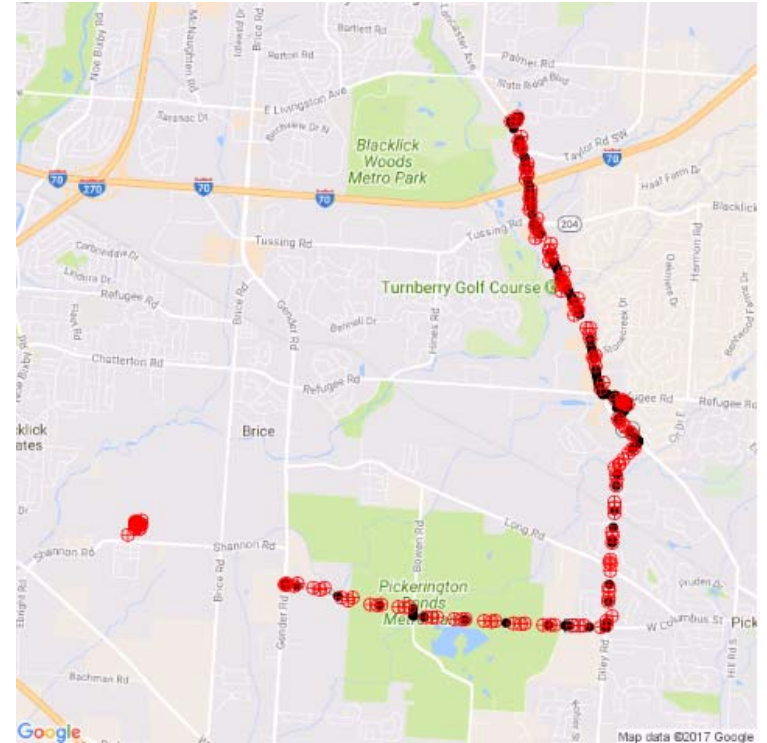
Pseudo-code for initial matching (*probabilistic spatial record linkage*)

1. Calculate “dwell” per user at each location (within ~100m radius) for rMove/Cuebiq
2. Sum dwell times per ~100m area and select top 3 “dwell” points (places where users spent a lot of time) for each user
3. Merge rMove/Cuebiq user-location tables by location and count number of matches user matches
4. Potential user matches are where 2 or more “common locations” align
5. Further refinement of user-matching algorithm (using both spatial and temporal distance) has been conducted & is being evaluated at this time

Maps of “Matched” User Locations



rMove locations
Cuebiq locations



- Locations on a Wednesday in April for “matched” users between ODOT study and Cuebiq Ohio data

On-going Analysis and Further Work

- How to assign probabilities to matches?
- How do we consider completeness/coverage of matches?
- Can we identify / characterize consistent patterns where rMove and Cuebiq traces differ?
- Can we document demographic biases at the disaggregate level?
- Can we verify the activity / trip duration bias at the disaggregate level?
- Can we devise better ways of expanding passive data using rMove traces?



Final Thoughts

Comparison of Expansion Methods

	Fix Trip Length Bias	Fix Coverage Problems	Fix Demographic Bias	Independent of Network	Ease of Application	Holdout Count Sample	Transparency
1 Market Penetration-based	✗	✓	✓✓	✓	✓	✓	-
2 Trip-Generation-based	✗	✓	✓✓	✓	-	✓	✓
3 Single-factor Scaling	✗	✗	✗	✓	✓	-	✓
4 Fratarling	✗	✓	✗	✓	✓	✗	✓
5 Iterative Screenlines	✓	✓	✓	✓	-	✓	✓
6 Direct ODME	✓	✓	✓	✗	✓	-	✗
7 Indirect ODME	✓	✓	✓	✗	✗	-	-
8 Parametric Scaling	✓✓	✓	✓	✗	✗	-	-
9 Disaggregate Trace Auditing	✓✓	✓	✓✓	✓	✗	✓✓	✓

- Ensemble methods best for now
- Count-based expansion necessary for now
- Disaggregate methods hold promise

What's Next?

- Data Driven Forecasting
 - Pivoting, destination choice models with constants
 - Better accuracy, analog to STOPS
- Accelerating Pace of Change
 - Transformational changes
 - Big data may provide key in more frequent updates
- Evolving Data & Methods
 - New data sources entering the market
 - Data fusion: surveys & big data



Contact

www.rsginc.com

Vince Bernardin, Jr, PhD
DIRECTOR OF TRAVEL FORECASTING

Vince.Bernardin@rsginc.com
812.200.2351